

ISRO successfully launches weather satellite INSAT-3DR

India's Space odyssey crossed a milestone when a locally-made cryogenic engine helped power a rocket that carried an advanced weather satellite to Space. The engine has been used only for testing purposes so far.



ISRO satellite INSAT-3DR, onboard GSLV-F05, launched from Sriharikota on September 8, 2016 (Image courtesy: www.thehindu.com)

Indian Space Research Organisation's (ISRO's) Geosynchronous Satellite Launch Vehicle-F05 (GSLV-F05) lifted off from Satish Dhawan Space Centre at Sriharikota in Andhra Pradesh on September 8, carrying with it INSAT-3DR weather satellite.

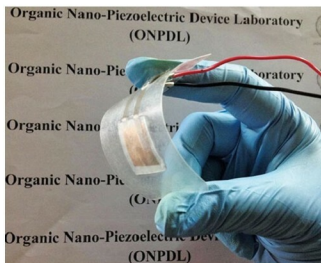
GSLV-F05 was the tenth flight of India's GSLV. It was the flight in which the indigenously-developed Cryogenic Upper Stage (CUS) was being carried onboard for the fourth time during a GSLV flight, as per a statement by ISRO, after the launch.

The 2211 kg rocket had three stages, and the last stage was fitted with the cryogenic engine, which provided a thrust 1.5 times greater than in vehicles propelled by conventional liquid rocket engines. After launch, the rocket's engines burnt and fell off in stages. At the last stage, the cryogenic engine came to life and took the satellite to Space. A propulsion put INSAT-3DR into a geostationary transfer orbit.

Biocompatible piezoelectric generator made from fish scales

Jadavpur University, Kolkata, physicists have used fish scales extracted from food waste to build tiny generators that can convert mechanical energy such as touch or sound vibrations to electrical energy. According to them, the piezoelectric device could be used to develop environmentally-friendly, self-powered electronics with a wide range of applications.

Piezoelectric materials respond to mechanical stress by separating positive and negative electrical charges, and therefore can be used to convert mechanical energy of vibrations into electrical energy. Piezoelectric generators that harvest energy from vibrations in the living environment would allow development of fully-independent, battery-free devices. These could be particularly useful for medical devices such as pacemakers and insulin pumps, and targeted drug-delivery systems that consume little power but need it continuously.



The battery-free generator could be a potential green power source for self-powered electronics (Image courtesy: www.physicsworld.com)